Minigraf 0507

control with Arduino Nano Every

from an Atari XE/XL computer (SIO) and from a PC (USB)



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1. What the interface does

The Aritma Minigraf A 0507 is a Czechoslovak pen plotter from the 1980s. It has no intelligence of its own: it only accepts a phase number for each stepper motor (X1, X2, X4 and Y1, Y2, Y4) and a pen signal. The Arduino interface does this work for the computer – it accepts readable commands (simplified HPGL), converts lines into motor steps and handles acceleration and deceleration.

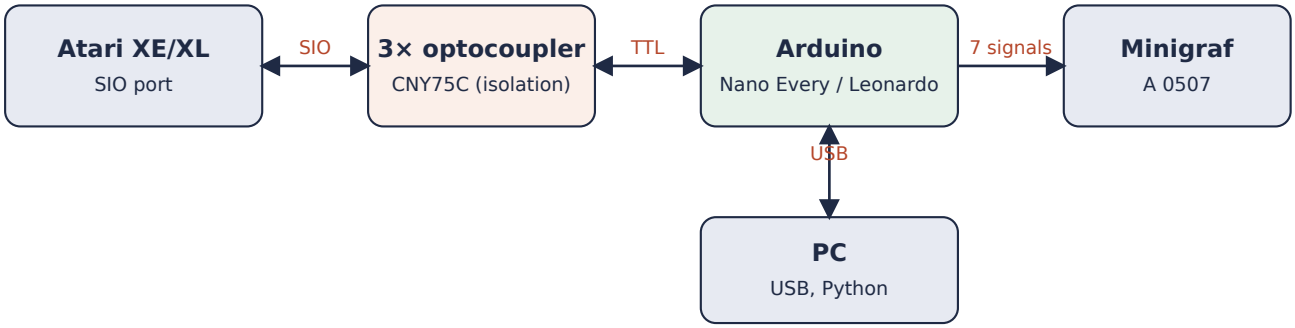


Fig. 1 – Overall setup

- **From the Atari** you draw with a plain `LPRINT` – the Arduino appears on the SIO bus as printer **P1:**.
- **From a PC** you draw with Python programs (`plot.py`, `text_plot.py`) or directly from the Arduino IDE Serial Monitor.
- The Atari is **galvanically isolated** by three optocouplers; the plotter connects to the Arduino directly.
- Transfer uses **handshaking**: the next command is sent only after the plotter has finished the previous one. Nothing gets lost.

2. Parts list

Part	Qty	Note
Arduino Nano Every (ATmega4809)	1	recommended; alternative Arduino Leonardo / Micro (chapter 3)
Optocoupler CNY75C (or CNY75A/B)	3	OK1 Data Out, OK2 Command, OK3 Data In
Resistor 1 kΩ	3	optocoupler LED current
Resistor 2.2 kΩ	2	pull-up on OK1 and OK2 outputs
Resistor 470 kΩ	3	base–emitter, speeds up transistor turn-off
SIO connector, 13 pin (or a cable from an SIO device)	1	pins 3, 4, 5, 7, 10
Minigraf connector, 12 pin, 2.54 mm pitch	1	key at position 7, bottom row of contacts only
Optional: 7× 220 Ω, 100 µF + 100 nF	–	signal protection, Arduino supply decoupling

Power

- **Arduino** is powered from USB (PC or USB adapter). Do not power it from the plotter.
- **Minigraf** has its own **24 V** supply (about 30 W, supply rated at least 1.5 A). Supply minus is also the logic ground – pins 1 and 11 of the data connector.
- **The Atari side** (optocoupler LEDs) is powered from SIO pin 10 (+5 V).

Important: Atari ground (GND_A) and Arduino/plotter ground (GND_D) must **never be connected**. That is what the optocouplers are for.

3. Supported Arduino boards

The firmware `minigraf_sio.ino` detects the board from the processor. Both boards have a separate hardware UART (`Serial1` on D0/D1) for SIO, so USB stays free for the PC and debugging. All six motor phases are always on one processor port and change at the same instant.

Signal	Nano Every	Leonardo / Micro	Minigraf pin	Function
SIO Data Out (via OK1)	`D0 (RX1)`	`D0 (RX1)`	–	data from Atari
SIO Data In (via OK3)	`D1 (TX1)`	`D1 (TX1)`	–	replies to Atari
SIO Command (via OK2)	`D2`	`D2`	–	start of SIO command
X1	`A0`	`A0`	`10`	X phase, bit 0
X2	`A1`	`A1`	`9`	X phase, bit 1
X4	`A2`	`A2`	`8`	X phase, bit 2
Y1	`A3`	`A3`	`4`	Y phase, bit 0
Y2	`A6`	`A4`	`3`	Y phase, bit 1
Y4	`A7`	`A5`	`2`	Y phase, bit 2
P (pen)	`D10`	`D10`	`5`	H = pen down
GND	`GND`	`GND`	`1, 11`	common ground with plotter

	Nano Every (recommended)	Leonardo / Micro
Processor	ATmega4809, 48 kB flash, 6 kB RAM	ATmega32U4, 32 kB flash, 2.5 kB RAM
Board package in Arduino IDE	Arduino megaAVR Boards (clones: MegaCoreX, Pinout "Nano Every")	Arduino AVR Boards (built in)
Motor phase port	PORTD (A0–A3, A6, A7)	PORTF (A0–A5)
Note	A6/A7 are full digital pins on the Nano Every	Micro uses the same pin labels

Minigraf connector

The cable connects to the plotter with the same type of edge connector used by ZX Spectrum bus peripherals, with **2.54 mm** pitch. It has only **12 positions**, and **position 7 is a key**. The plotter PCB has contacts on the bottom side only, so **only the bottom row** of the connector is used.



Fig. – Connector position numbering (key at position 7 shown in black)



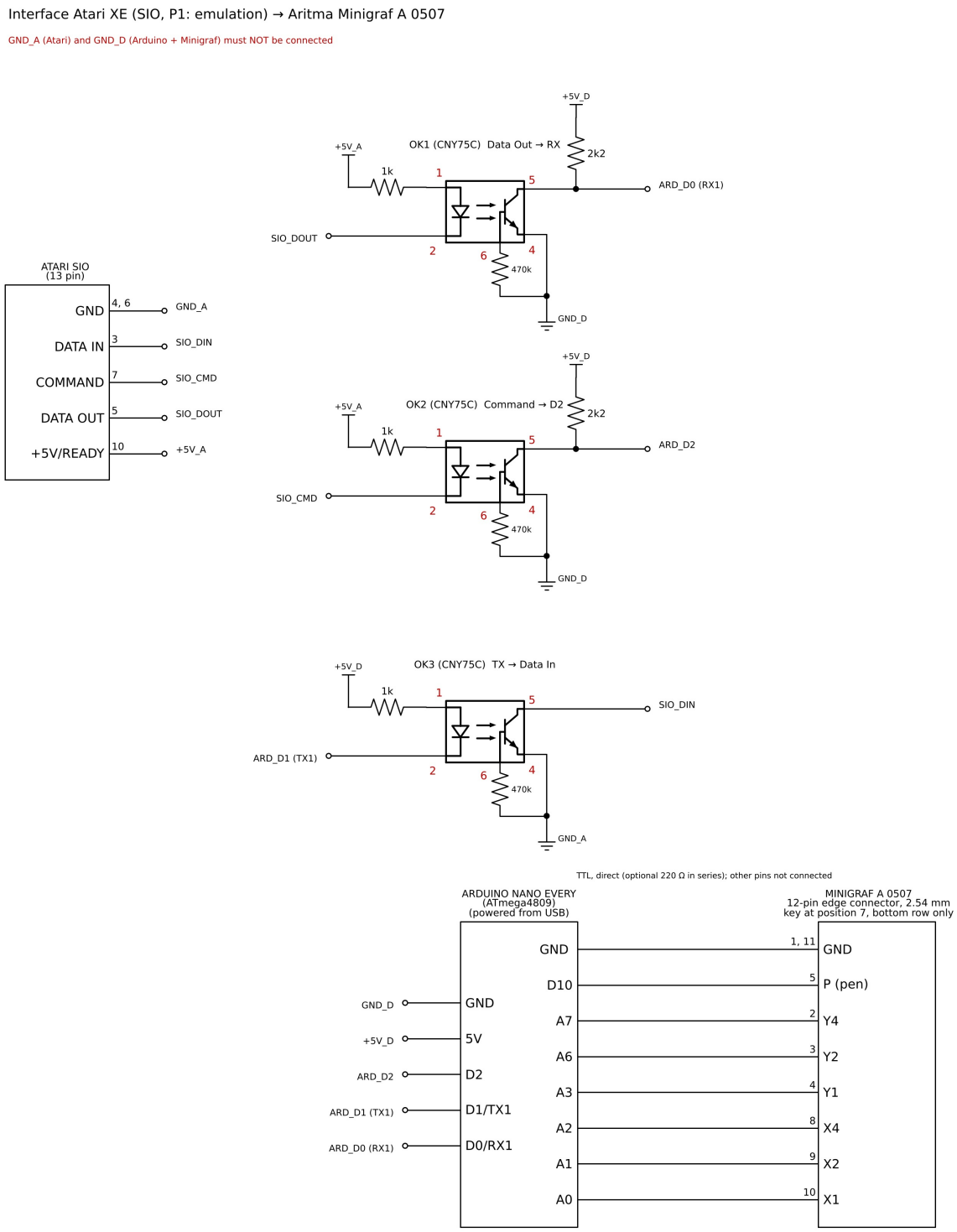
	12	11	10	9	8	7	6	5	4	3	2	1
signal:	**nc**	**GND**	**X1**	**X2**	**X4**	*key*	**nc**	**P**	**Y1**	**Y2**	**Y4**	**GND**
Nano Every:		GND	A0	A1	A2			D10	A3	A6	A7	GND

bottom row of plotter contacts (position 7 = key, no contact)

Fig. – Signals at each position and the matching Arduino Nano Every pins

> **Caution:** the key **counts as pin 7** in the numbering even though it has no contact. Contacts to the left of the key are therefore 8 to 12, not 7 to 11. Numbers moulded on some connectors do not match this numbering – check with an ohmmeter that pins 1 and 11 are plotter ground before connecting.

4. Schematic

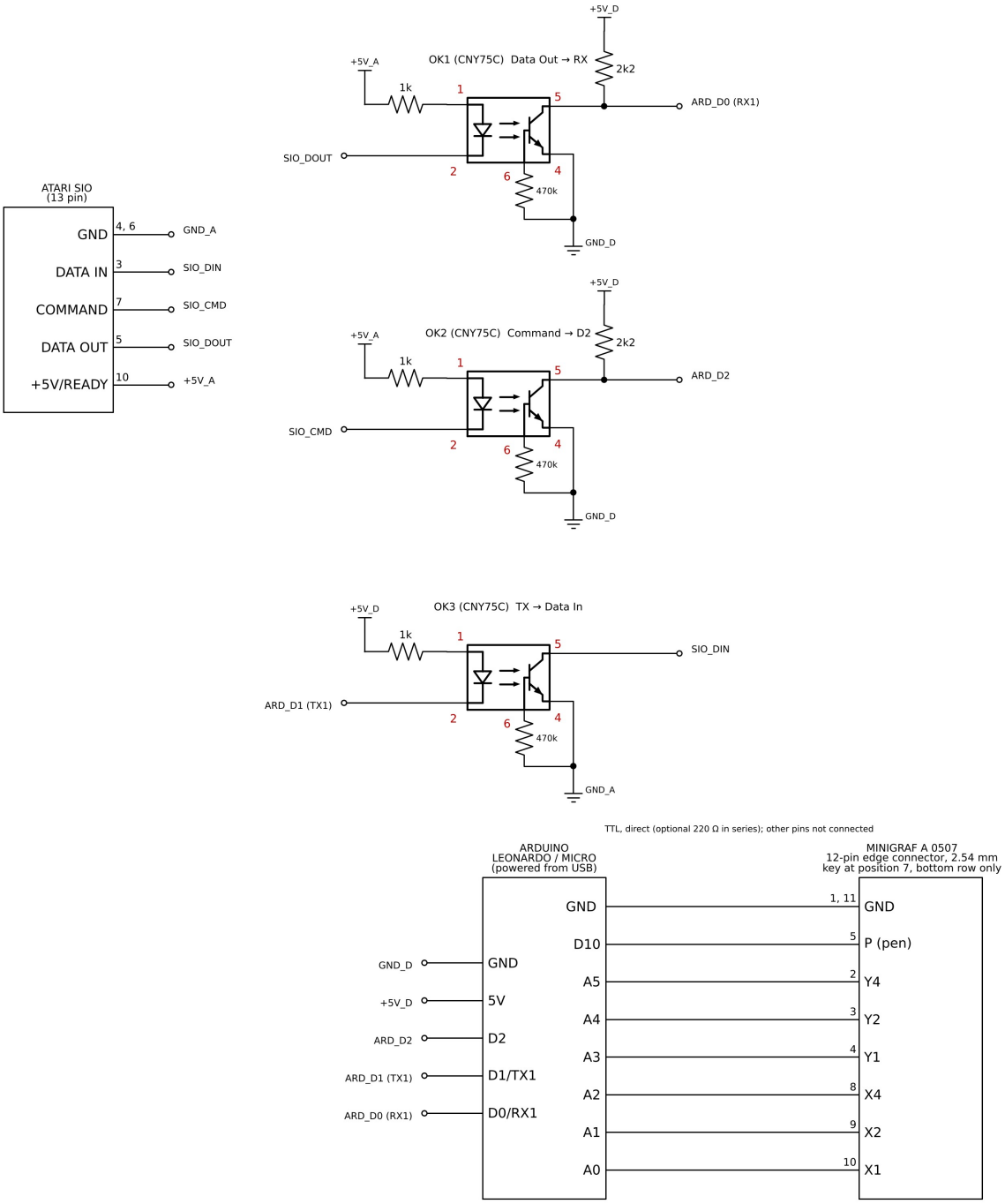


Note: +5V_A = SIO pin 10 (Atari), +5V_D = Arduino 5 V. All optocouplers are non-inverting.
If D0 is not below 0.8 V while the Atari transmits, reduce 1k at OK1/OK2 to 680 Ω.

Fig. 2 – Wiring with Arduino Nano Every

Interface Atari XE (SIO, P1: emulation) → Aritma Minigraf A 0507

GND_A (Atari) and GND_D (Arduino + Minigraf) must NOT be connected



Note: +5V_A = SIO pin 10 (Atari), +5V_D = Arduino 5 V. All optocouplers are non-inverting.
If D0 is not below 0.8 V while the Atari transmits, reduce 1k at OK1/OK2 to 680 Ω.

Fig. 3 – Alternative wiring with Arduino Leonardo / Micro (only Y2 = A4 and Y4 = A5 differ)

5. Software installation

Arduino firmware

- In the Arduino IDE open the **Boards Manager**, search for **megaAVR** and install **Arduino megaAVR Boards**.
- Select **Tools → Board → Arduino megaAVR Boards → Arduino Nano Every** and the correct port.
- Open `minigraf_sio.ino` and upload it.
- The message `jtagmkII_initialize(): Cannot locate "flash" and "boot" memories` is normal and harmless on the Nano Every.
- Nano Every clone with a CH340 USB chip: use **MegaCoreX** instead, board ATmega4809 and **Pinout: Nano Every**.

Python on the PC (Windows)

- Install Python 3 and run in the **command prompt** (not in the Python window with >>>):

```
py -m pip install pyserial
py -m pip install matplotlib      (only for --preview)
```

- Put `plot.py` and `text_plot.py` in the same folder, e.g. `C:\plotter`.
- The default port is **COM4**. Give another port before the file name or with `--port`.
- The **Serial Monitor must be closed** while plotting from Python – only one program can open the port.

6. Coordinates and plotting area

Parameter	Value
Unit	1 step = 0.125 mm (8 steps = 1 mm)
X axis (short side)	0 – 1503 steps (188 mm)
Y axis (long side)	0 – 2103 steps (263 mm)
Point 0,0	where the head is when the Arduino starts or when IN is sent
Centre of area	X = 752, Y = 1052
Position on A4	point 0,0 about 11 mm from the short and 17 mm from the long paper edge

The plotter has no limit switches – the Arduino knows the position only by counting steps. Before each drawing move the head by hand to the bottom-left corner and send `IN;` (the PC programs do this automatically).

7. Command reference

Commands are the same from the Atari, the Serial Monitor and in PC files. Each command starts with two letters, numbers are separated by commas and commands by semicolons. Several commands may be on one line.

Command	Meaning	Example
`IN;`	initialise: current position = 0,0, pen up, absolute coordinates	`IN;`
`PU x,y;`	pen up and move to x,y (several points allowed)	`PU752,1052;`
`PD x,y,...;`	pen down and draw through the given points	`PD800,100,800,900;`
`PA;`	following coordinates are absolute (default)	`PA;PU0,0;`
`PR dx,dy;`	following coordinates are relative to current position	`PR80,0;`
`CI r[,n];`	circle of radius r around current position, n segments (default 72)	`PU750,1050;CI400;`
`VS v;`	speed in mm/s; VS; = default (60 mm/s)	`VS40;`
`PT d,u;`	pen delay in ms after pen down / pen up; PT; = default (0 / 70)	`PT20,70;`
`?`	Serial Monitor only: prints status (uptime, position, pen, SIO message count)	`?`

Useful moves

What	Command
Head back to origin	`PU0,0;`
Head to centre of page	`PU752,1052;`
Move 10 mm right (without drawing)	`PU;PR80,0;PA;`
Square 50 × 50 mm	`PU200,200;PD600,200,600,600,200,600,200,200;PU;`
New origin at current position	`IN;`

Firmware defaults

Constant in minigraf_sio.ino	Value	Meaning
`STEP_US`	`2083`	speed 60 mm/s (= 125000 / speed)
`START_US / RAMP_US`	`9000 / 150`	gradual start and stop of every line
`MIN_STEP_US`	`1500`	maximum allowed speed about 83 mm/s
`PEN_DOWN_MS / PEN_UP_MS`	`0 / 70`	pen delay (tested with a 0.5 mm marker)
`PEN_DOWN_LEVEL`	`HIGH`	P signal level for pen down
`X_DIR / Y_DIR`	`+1 / +1`	change to −1 to reverse an axis
`PRN_TIMEOUT_S`	`255`	how long the Atari waits for a line to finish

8. Plotting from the Atari

On the SIO bus the Arduino behaves as printer **P1:**. You draw with a plain LPRINT in Atari BASIC (or by writing to device `P:` from other programs). After each line the Atari waits until the Arduino confirms the drawing is finished – no timing needed.

Example Atari BASIC program

```
10 REM MINIGRAF 0507 - DEMO
20 LPRINT "IN;"
30 REM FRAME
40 LPRINT "PU100,100;PD1400,100,1400,2000,100,2000,100,100;"
50 REM CIRCLE IN THE MIDDLE
60 LPRINT "PU750,1050;CI400;"
70 REM VERTICAL LINES
80 FOR I=0 TO 20
90 LPRINT "PU";100+I*60;",300;PD";100+I*60;",900;"
100 NEXT I
110 LPRINT "PU0,0;"
```

Quick commands from the Atari

Atari BASIC	Result
`LPRINT "PU0,0;"`	head back to origin
`LPRINT "VS40;"`	slower drawing (40 mm/s)
`LPRINT "VS;PT;"`	back to default speed and pen delay
`LPRINT "PU750,1050;CI200;"`	circle 50 mm in diameter in the middle

Atari tips

- One LPRINT may contain several commands; longer lines are split into several SIO records automatically.
- Atari BASIC prints numbers without spaces, so `LPRINT "PU";X;",";Y;";"` works directly.
- If a single line took longer than about 4 minutes, the Atari would report error 138 – split such a line into several LPRINT.
- While commissioning, the Serial Monitor shows `SIO: status` and `SIO: zapis OK` for each LPRINT.

9. Plotting from a PC

9.1 Serial Monitor (manual commands)

In the Arduino IDE open the Serial Monitor, set **115200 baud** and **Newline**. Each line you type is drawn and the Arduino replies `hotovo` ("done"). Board LED blinking = program running, steady = drawing.

9.2 Plotting an HPGL file – plot.py

```
py plot.py drawing.hpgl              (HPGL from Inkscape, port COM4)
py plot.py napis.hpgl --scale 1      (file already in plotter steps)
py plot.py COM7 drawing.hpgl         (another port)
py plot.py drawing.hpgl --dry        (only print what would be sent)
```

Option	Meaning
`--scale 0.2`	coordinate scale; 0.2 = HPGL from Inkscape (40 units/mm), 1 = plotter steps
`--speed 40`	speed in mm/s for this drawing
`--pen 0,70`	pen delay in ms (after pen down, after pen up)
`--dry`	send nothing, only print the converted commands

Plotting from Inkscape: File → Save As → *HP Graphics Language (*.hpgl)*, resolution 1016 dpi, leave "Center zero point" unticked.

9.3 Lettering and text – text_plot.py

The program has its own stroke font (upper and lower case, digits, punctuation, Czech diacritics). Three modes:

```
Large lettering across the whole width (A4 landscape):
  py text_plot.py "HELLO" --frame --star --bold 3 --send

Normal text in lines, wrapped automatically ( | = new line):
  py text_plot.py "Hello,|this is Minigraf." --size 5 --send
  py text_plot.py --file letter.txt --size 4 --portrait --send

Test page (6 / 4 / 3 / 2 mm):
  py text_plot.py --test --portrait --send
```

Option	Meaning
`--send`	send to the plotter right away (otherwise only creates napis.hpgl)
`--size 5`	capital letter height in mm – normal text mode
`--portrait`	A4 portrait like a printer (lines along the short side)
`--file letter.txt`	text from a file (save as UTF-8)
`--frame / --star`	double frame / eight-pointed star above large lettering
`--bold 1–4`	bold: letters are traced several times with a small offset
`--margin 8`	text margin in mm (default 15 for lettering, 10 for text)
`--flip`	rotate the drawing by 180° if it comes out upside down
`--preview`	save a .png preview (needs matplotlib)
`--speed / --pen / --port`	same as plot.py

Recommended letter size by pen

Pen line width	Smallest well-readable letters
0.5 mm	about 5 mm
0.3 mm	about 3 mm
0.1–0.2 mm	about 2 mm

10. Example results



Fig. 4 – Large lettering with frame and star, speed 80 mm/s, 0.5 mm marker.
`py text_plot.py "ZDENĚK 0507" --frame --star --bold 3 --send --speed 80`

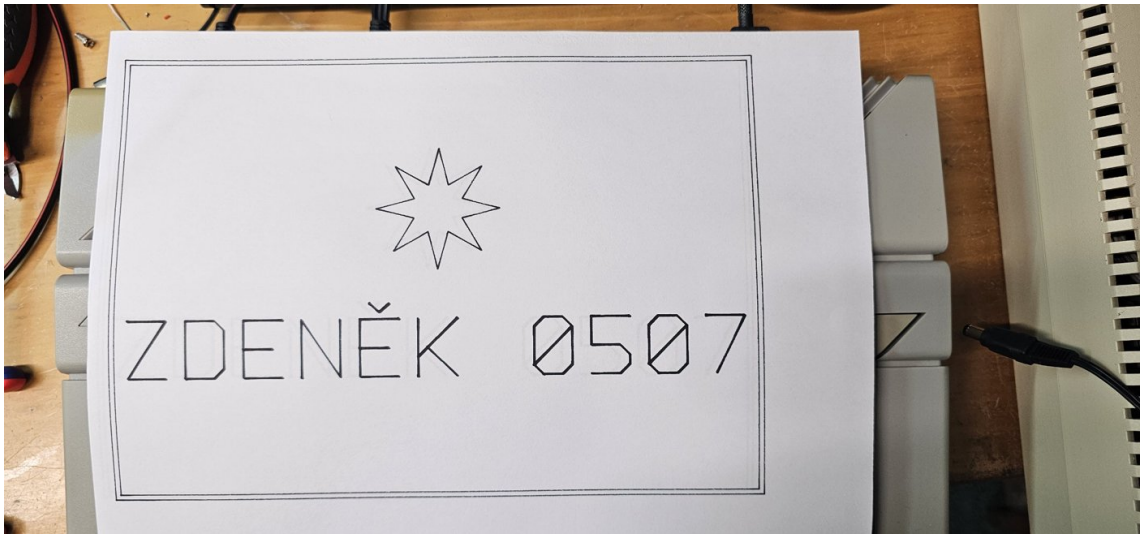


Fig. 5 – The same drawing from the first tests, view of the plotter.

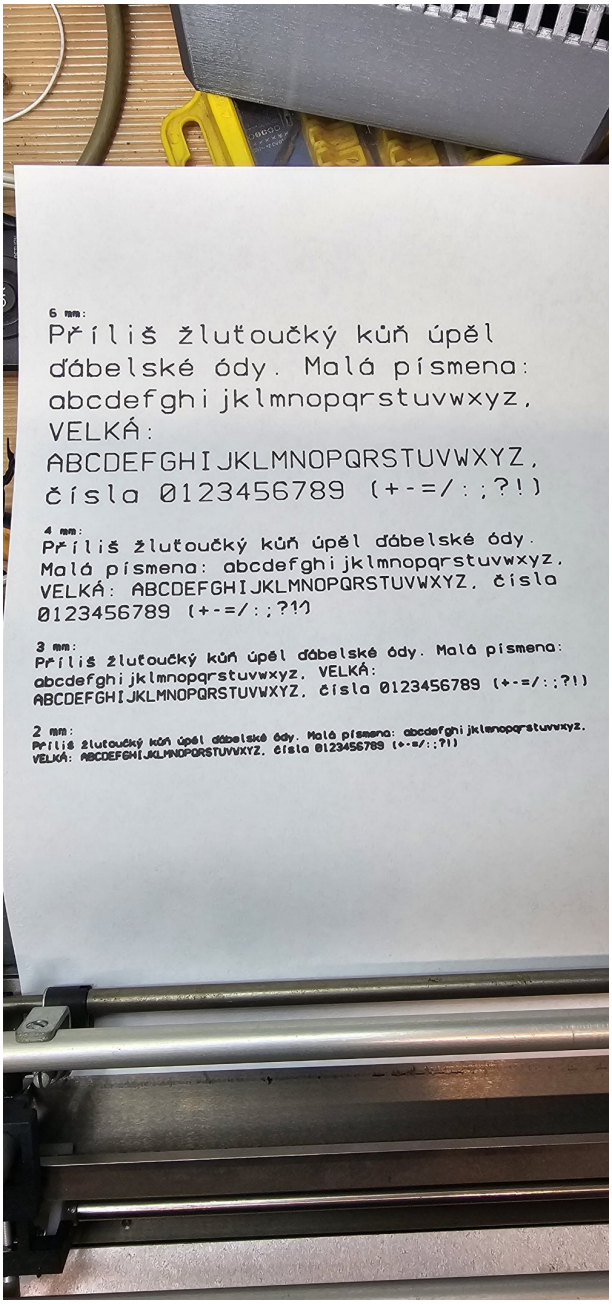


Fig. 6 – Test page

`py text_plot.py --test --portrait --send`

Default settings: speed 60 mm/s, pen delay 0 / 70 ms, 0.5 mm marker.

- 6, 4 and 3 mm: clean and easy to read, loops close, diacritics in place.
- 2 mm: readable, but a 0.5 mm marker is at its limit – letters run together. A 0.3 mm marker will be clean.
- Lines rise slightly to the right: the paper was loaded slightly skewed. Align the paper edge with the roller before plotting.

11. Troubleshooting

Symptom	Cause and fix
Head stops and hums	motor is losing steps – lower the speed (--speed or VS), check the 24 V supply and mechanics
plot.py stuck at a percentage	Arduino not responding; the current plot.py reports "ARDUINO SE RESTARTOVALO" (Arduino restarted). Fix: 100 µF + 100 nF on Arduino 5 V, shorter ground to the plotter, short USB cable
could not open port / Access denied	Serial Monitor open or wrong port number
No module named serial	py -m pip install pyserial
Nothing happens from the Serial Monitor	set "Newline" and 115200 baud (bottom right)
Blobs at the start of lines	shorten delay after pen down: --pen 0,70 or PT0,70;
Start of line missing	lengthen delay after pen down: --pen 30,70
Tails or smudges between letters	lengthen delay after pen up: --pen 0,100
Drawing mirrored	in minigraf_sio.ino change X_DIR or Y_DIR to −1
Drawing upside down	option --flip
Pen does the opposite (PD lifts)	change PEN_DOWN_LEVEL to LOW
Nothing after LPRINT, monitor silent	check OK1 and OK2; at rest the status must show Command=H
"chyba souctu" (checksum error) in monitor	slow edges at OK1 – reduce 1 kΩ to 680 Ω
Atari reports error 138	Arduino not responding or one line takes longer than the timeout – split the line
Text lines are slanted	paper loaded skewed – align it with the roller
Marker smears even mid-line	squashed tip – replace the marker

12. Project files

File	Content
`firmware/minigraf_sio/minigraf_sio.ino`	firmware for Arduino Nano Every and Leonardo/Micro
`pc/plot.py`	sending HPGL files from a PC
`pc/text_plot.py`	lettering, text in lines and test page
`docs/`	manuals (CZ, EN), schematics and photos

Note: the firmware and PC programs print their messages in Czech. Connector pinout based on cygnus.speccy.cz (Aritma Minigraf 0507); signal meaning from the MZXR ZX Spectrum documentation and the MAME emulator.